Exact Encounter Analysis for Two Lazy Reflecting Walkers

Conclusion

For the current finite one-dimensional reflecting lattice with synchronous lazy updates and co-location-only absorption, no robust F_2 double peak was found. The unusual curves seen across Stage 1, Stage 2, and Stage 2b are best explained as parity artifacts, same-target long tails, or weak target-shift shoulders. In particular, the representative boundary cases A–C are not double peaks and are not classified as boundary-induced delayed channels; the best retained target-shift case is only a weak shoulder, not a double peak.

The visible bumps or shoulders occur in specific regimes rather than as a universal double-peak mechanism. Short-distance, strongly asymmetric mobility cases can show jagged raw peaks from odd/even parity; the negative control

$$N = 21, \qquad (a, b) = (6, 2), \qquad q_1 = 0.02, \qquad q_2 = 0.90$$

is the clearest example. Slow near-boundary cases A–C show long tails or shoulders, but the dominant encounter site is unchanged between early and late windows: $4 \to 4$, $6 \to 6$, and $3 \to 3$. The best retained target-shift case

$$N = 31, \qquad (a, b) = (2, 4), \qquad q_1 = 0.70, \qquad q_2 = 0.20$$

has a weak site shift $4 \to 3$, but still no true F_2 double peak. Thus the jagged figure is normal for this synchronous discrete-time model: it is a parity-comb artifact in raw $f(t)$. The decision curve is F_2, which pairs odd/even times before applying the repository's peak-valley rule.

Model and Exact Absorbing-Chain Formulation

Sites are $1, \dots, N$. Walker i has mobility $q_i \in (0, 1]$. At an interior site, the walker moves left/right with probabilities $q_i/2$ and stays with probability $1 - q_i$. At the reflecting boundary, attempted outside motion is converted into staying:

$$P_i(1, 1) = 1 - q_i/2, \quad P_i(1, 2) = q_i/2, \qquad P_i(N, N) = 1 - q_i/2, \quad P_i(N, N-1) = q_i/2.$$

The synchronous independent update is

$$P = P_1 \otimes P_2.$$

The transient set is the off-diagonal state space

$$O = \{(x, y) : x \neq y\},$$

and the absorbing encounter set is

$$D = \{(k, k) : 1 \leq k \leq N\}.$$

With $Q = P[O, O]$ and $R = P[O, D]$, the exact encounter-position distribution is

$$f_k(t) = \alpha Q^{t-1} R_{\cdot,k}, \qquad f(t) = \sum_k f_k(t).$$

All reported curves and identities are exact sparse Markov-chain calculations, not Monte Carlo estimates.

Spatial Configuration Views

Figure 1 shows the actual spatial configuration for each selected case. The left panel in each row is the physical reflecting one-dimensional lattice: the two walker starts are marked, the reflecting walls are drawn at sites 1 and N, and the dominant early/late encounter sites are shown directly on the lattice. The right panel in each row is the ordered Markov state space (x, y). Its diagonal is the absorbing co-location set $D = \{(k, k)\}$; the start is a single off-diagonal point.

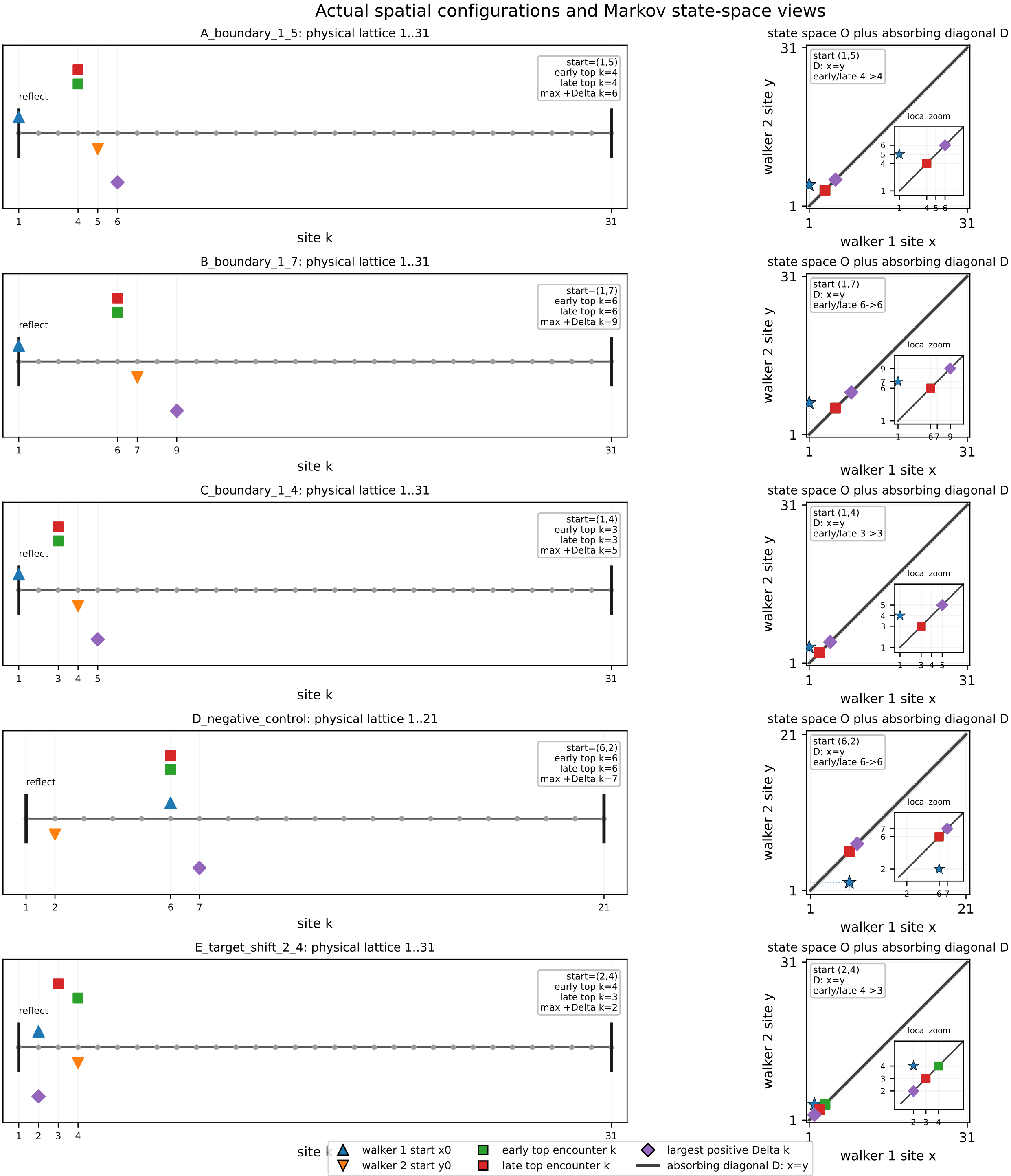

Figure 1: Actual spatial configurations. Left: physical 1D lattice. Right: ordered state space (x, y), where the diagonal is the absorbing encounter set. Marker meanings are shown in the legend: blue up-triangle is walker 1 start x_0, orange down-triangle is walker 2 start y_0, green square is early top encounter site, red square is late top encounter site, and purple diamond is the site with largest positive $\Delta_k = C_{\mathrm{late}} - C_{\mathrm{early}}$. Each right panel also includes a local zoom near the start and dominant encounter sites.

Operational Double-Peak Rule and Figure Reading

A raw second bump is not enough. The final rule first forms the parity-pair curve

$$F_2(n) = f(2n-1) + f(2n),$$

which removes odd/even synchronous-update oscillations. A case is called a robust F_2 double peak only if all of the following hold:

1. F_2 has a main local maximum and a later true local maximum.
2. The later maximum is more than two F_2 bins after the main peak.
3. The later peak height is at least 5% of the main peak height.
4. The valley between the two peaks is at most 80% of the smaller peak height.

If these conditions fail, the feature is reported as a shoulder, long tail, target-shift shoulder, or artifact. This is why cases A–C and the best target-shift case are not called double peaks.

This is aligned with the repository's existing visual double-peak convention, $\rho = h_v/\min(h_1, h_2) \le 0.8$. The encounter-specific part is only the preprocessing step: synchronous co-location curves can alternate strongly between odd and even time steps, so the final double-peak claim applies the same peak-valley rule to F_2, not directly to raw $f(t)$.

In the diagnostic figures, dashed green/red vertical lines mark the main and late diagnostic times. The C_k panel compares normalized encounter-site mass in the early and late windows. The heatmap is $\log_{10} f_k(t)$, so it is meant to show which encounter sites carry mass at each time rather than to show a probability density on a linear colour scale. Light gray bands and inset axes mark local zoom windows for the jagged, shoulder, or F_2-decision regions that are hard to inspect on the full time axis.

Formula and Green-Function Validation

The fundamental-matrix identities are

$$\mathbb{E}[T] = \alpha(I-Q)^{-1}\mathbf{1}, \qquad p_k = \alpha(I-Q)^{-1}R_{\cdot,k},$$

and

$$\tau_k = \frac{\alpha(I-Q)^{-1}(I-Q)^{-1}R_{\cdot,k}}{p_k}.$$

The validation table shows numerical closure of mass, mean, encounter-position probabilities, and the $p_k\tau_k$ decomposition.

Table 1: Formula validation. Errors are at floating-point scale.

case	N	$\sum_t f(t)$	tail mass	mean error	$p\tau$ mean error
L5 edge q04 q09	5	1.000000	9.42×10^{-13}	2.50×10^{-10}	8.88×10^{-15}
L6 inner q07 q03	6	1.000000	9.78×10^{-13}	4.58×10^{-10}	2.13×10^{-14}
L7 asym q02 q08	7	1.000000	9.63×10^{-13}	6.25×10^{-10}	4.97×10^{-14}

For $N = 5$, start $(1, 5)$, $q_1 = 0.4$, $q_2 = 0.9$, the absorbing-chain probability-generating function agrees with the determinant Green-function formula. The maximum absolute error over $z = 0.2, 0.5, 0.8, 0.95$ is 3.52×10^{-16}.

Table 2: Green-function determinant validation.

z	absorbing PGF	determinant PGF	absolute error
0.20	0.000789	0.000789	3.52×10^{-16}
0.50	0.015682	0.015682	5.20×10^{-17}
0.80	0.150581	0.150581	2.78×10^{-16}
0.95	0.562263	0.562263	1.11×10^{-16}

Mean Encounter Time Versus Mobility Ratio

The mean-ratio pilot shows that an interior maximum of the exact mean encounter time is not universal. Two starts have an interior maximum over the scanned range; the boundary-to-boundary fixed-sum case has an interior minimum and an endpoint maximum.

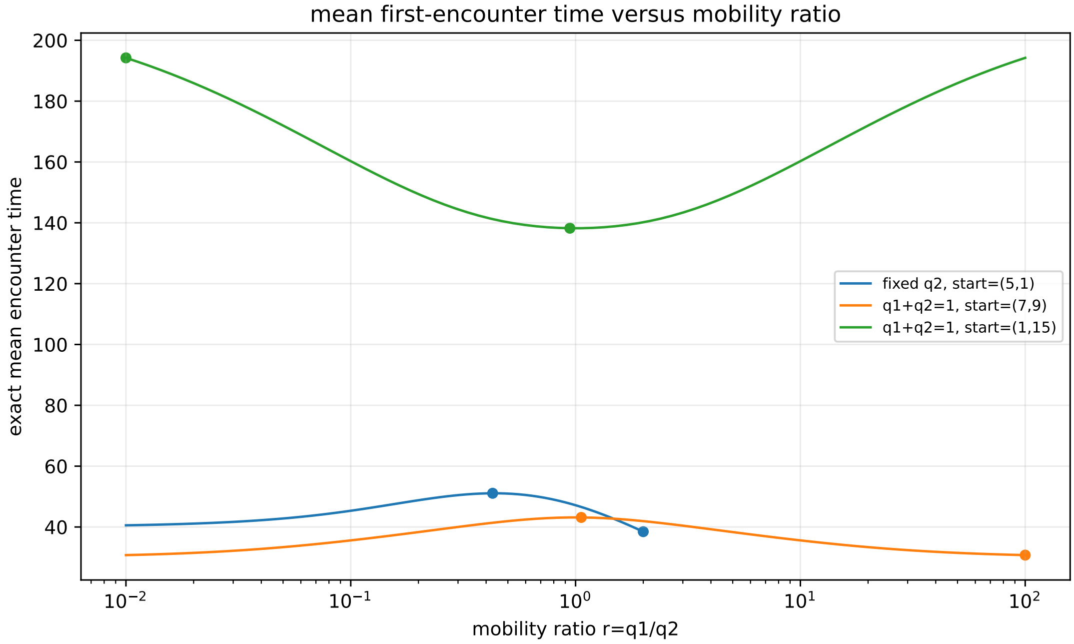

Figure 2: Exact mean encounter time as a function of the mobility ratio $r = q_1/q_2$ for the three pilot scans.

Table 3: Mean-ratio extrema.

scan	extremum	r	q_1	mean
fixed q_2, start $(5, 1)$	global/local max	0.427671	0.213836	51.0852
fixed q_2, start $(5, 1)$	global min	2.000000	1.000000	38.4629
fixed sum, start $(7, 9)$	global/local max	1.060026	0.514569	43.1421
fixed sum, start $(7, 9)$	global min	100.000000	0.990099	30.7548
fixed sum, start $(1, 15)$	global max	0.010000	0.009901	194.223
fixed sum, start $(1, 15)$	global/local min	0.943373	0.485431	138.226

Why the Stage-1 Double Peaks Were Artifacts

Stage 1 ranked raw curve shapes. This promoted short-distance parity combs: raw $f(t)$ has repeated alternating local maxima, but the parity-pair curve

$$F_2(n) = f(2n-1) + f(2n)$$

collapses the pattern to a single meaningful peak. These rows are therefore rejected as parity or ballistic-short-distance artifacts.

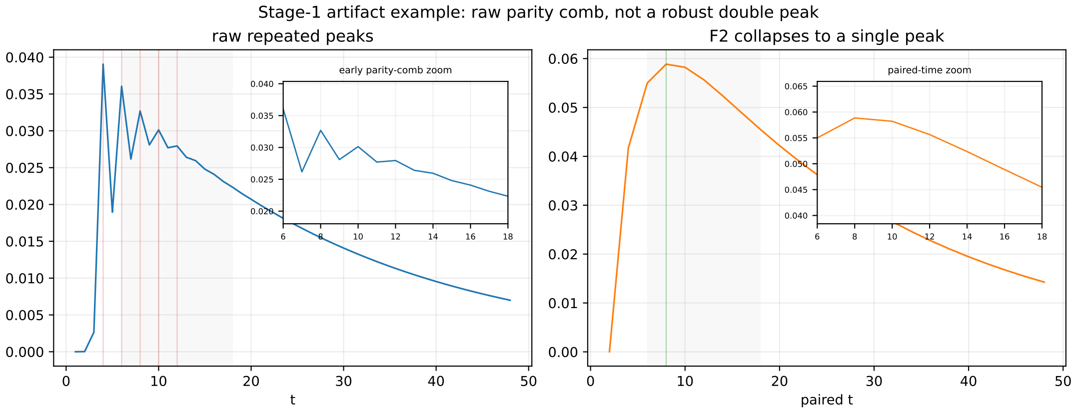

Figure 3: Stage-1 artifact example. Multiple raw peaks disappear after pairing adjacent odd/even times into F_2. Insets enlarge the early parity-comb region and the paired-time decision region.

Stage-2 and Stage-2b: No Robust F_2 Double Peaks

Stage 2 introduced smoothing persistence, parity-pair filtering, and negative controls. Stage 2b added mechanism-focused checks: boundary-null controls, target-shift reranking, encounter-site windows, grouped $f_G(t)$ curves, and leading eigenmodes of Q. In the Stage-2b case table plus the top-30 target-shift table, the number of robust F_2 double peaks is zero.

Table 4: Final mechanism classification for the representative cases.

case	class	N	start	q_1	q_2	top k early/late
A boundary 1–5	same-target tail	31	$(1,5)$	0.05	0.02	$4 \to 4$
B boundary 1–7	same-target tail	31	$(1,7)$	0.10	0.02	$6 \to 6$
C boundary 1–4	same-target tail	31	$(1,4)$	0.05	0.02	$3 \to 3$
D negative control	parity artifact	21	$(6,2)$	0.02	0.90	$6 \to 6$
best target shift	weak target-shift shoulder	31	$(2,4)$	0.70	0.20	$4 \to 3$

Same-Target Long-Tail Cases A–C

Cases A–C are shoulders or long tails, not double peaks. Their early and late encounter-site distributions have the same dominant k, and the translated interior controls do not show a large boundary-specific amplification. They are therefore not called boundary-induced delayed channels.

Weak Target-Shift Shoulder

The clearest retained target-shift case is

$$N = 31, \qquad (a,b) = (2,4), \qquad q_1 = 0.7, \qquad q_2 = 0.2.$$

Its early/late encounter-site distributions shift from top site $k=4$ to $k=3$, with $JSD = 0.0537$, maximum positive Δ_k at $k=2$, and target-shift score 1.43×10^{-3}. This is a weak target-shift shoulder, not a double peak: F_2 still lacks two true local maxima separated by a visible valley.

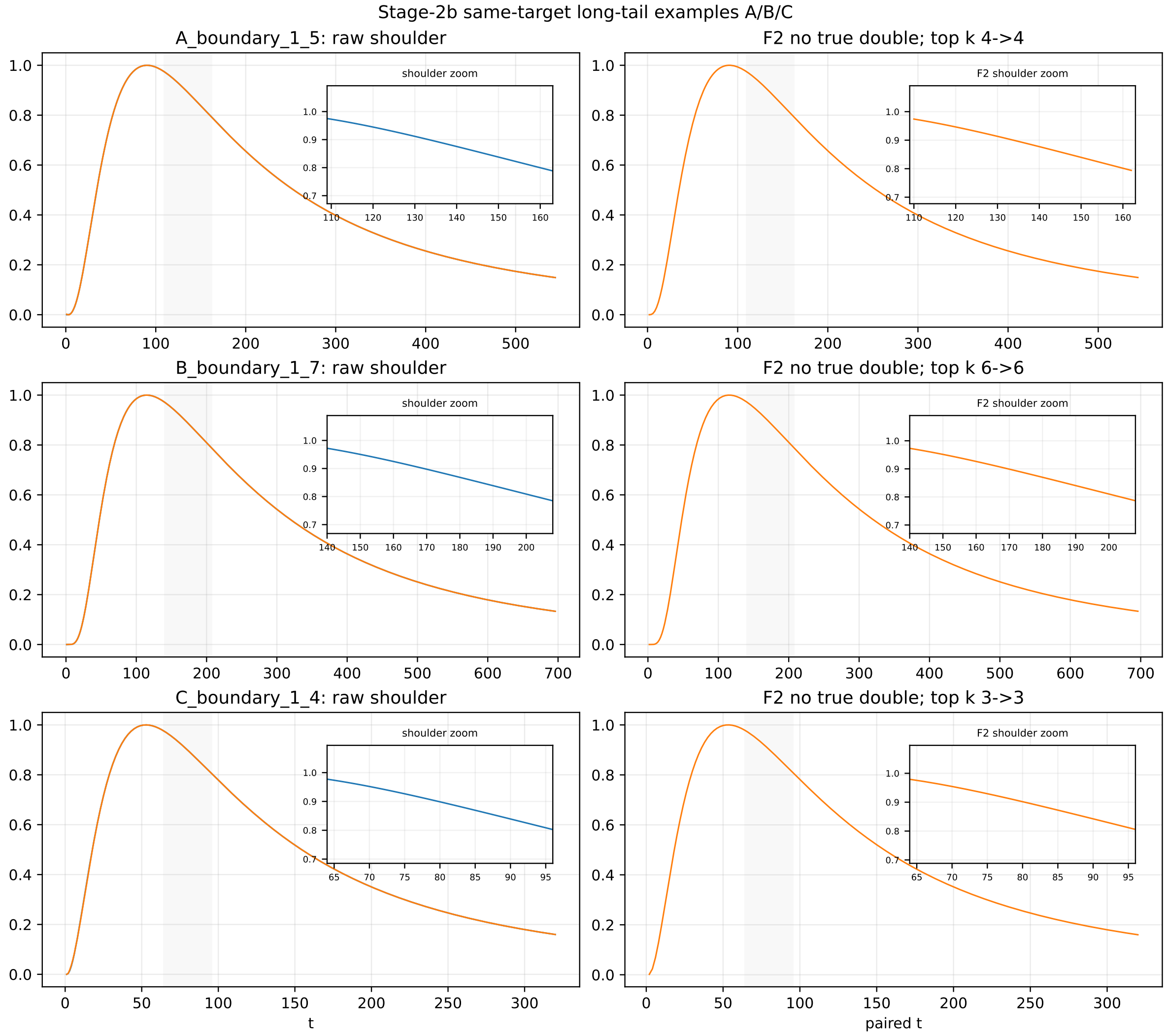

Figure 4: Same-target long-tail examples A–C. The raw curves have late shoulders, but F_2 has no true second local maximum with a visible valley. Insets zoom the shoulder windows.

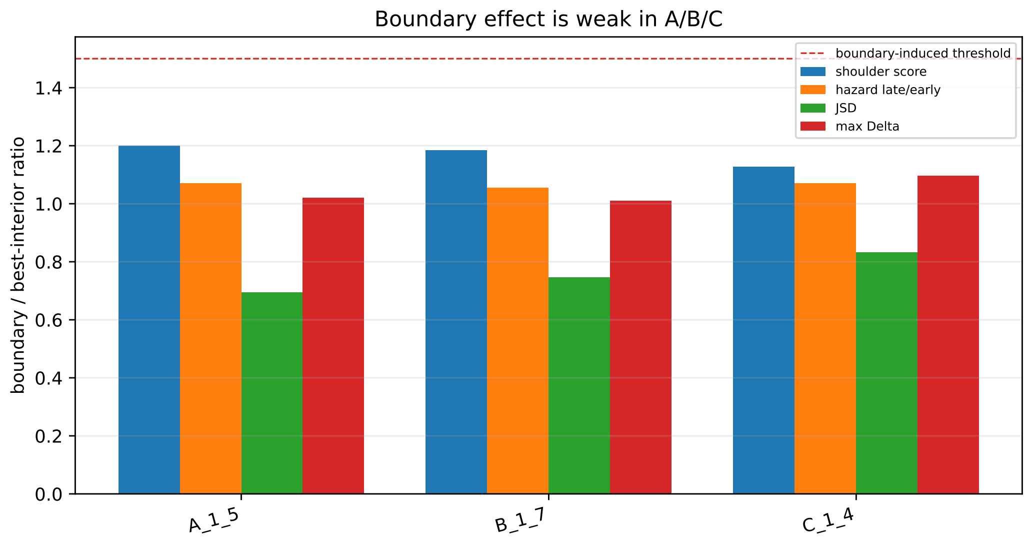

Figure 5: Boundary-null comparison. The boundary cases do not dominate their translated interior controls strongly enough to support a boundary-induced mechanism.

Figure 6: Best target-shift example. The C_k distribution shifts, but the total curve does not become a robust F_2 double peak. Insets zoom the weak shoulder and F_2-decision windows.

Spectral Interpretation

The leading eigenvalues of Q are close to one, and the late curves are consistent with slow-mode relaxation rather than a separate second-peak mechanism. For case A, the leading mode has $\lambda = 0.99982753$ and top R-channel $k = 16$, while the early and late dominant encounter site remains $k = 4$. The target-shift example shows a modest C_k redistribution, but not a distinct slow channel that creates a true second peak.

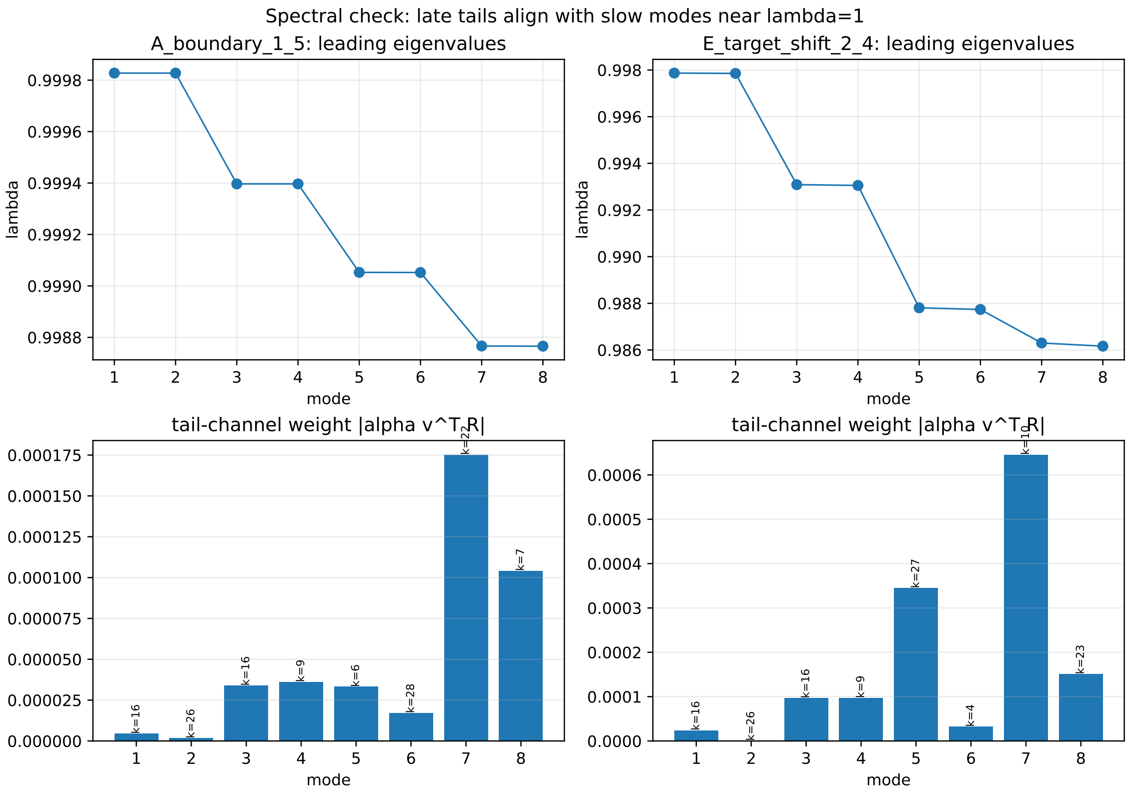

Figure 7: Leading eigenmodes for case A and the best target-shift shoulder.

Diagnostic Appendix

The appendix figures show, for each selected case, raw $f(t)$, smoothed $f(t)$, F_2, $\log f(t)$, hazard $h(t) = f(t)/S(t-1)$, log-derivative of F_2, k-time heatmap, and early/late C_k.

Reproducibility

The final report is regenerated from the existing Stage 1, Stage 2, and Stage 2b outputs by

```
PYTHONPATH=research/reports/final_multitimescale_fpt_encounter/code \
  .local/encounter-py311/bin/python -m encounter_search.finalize
```

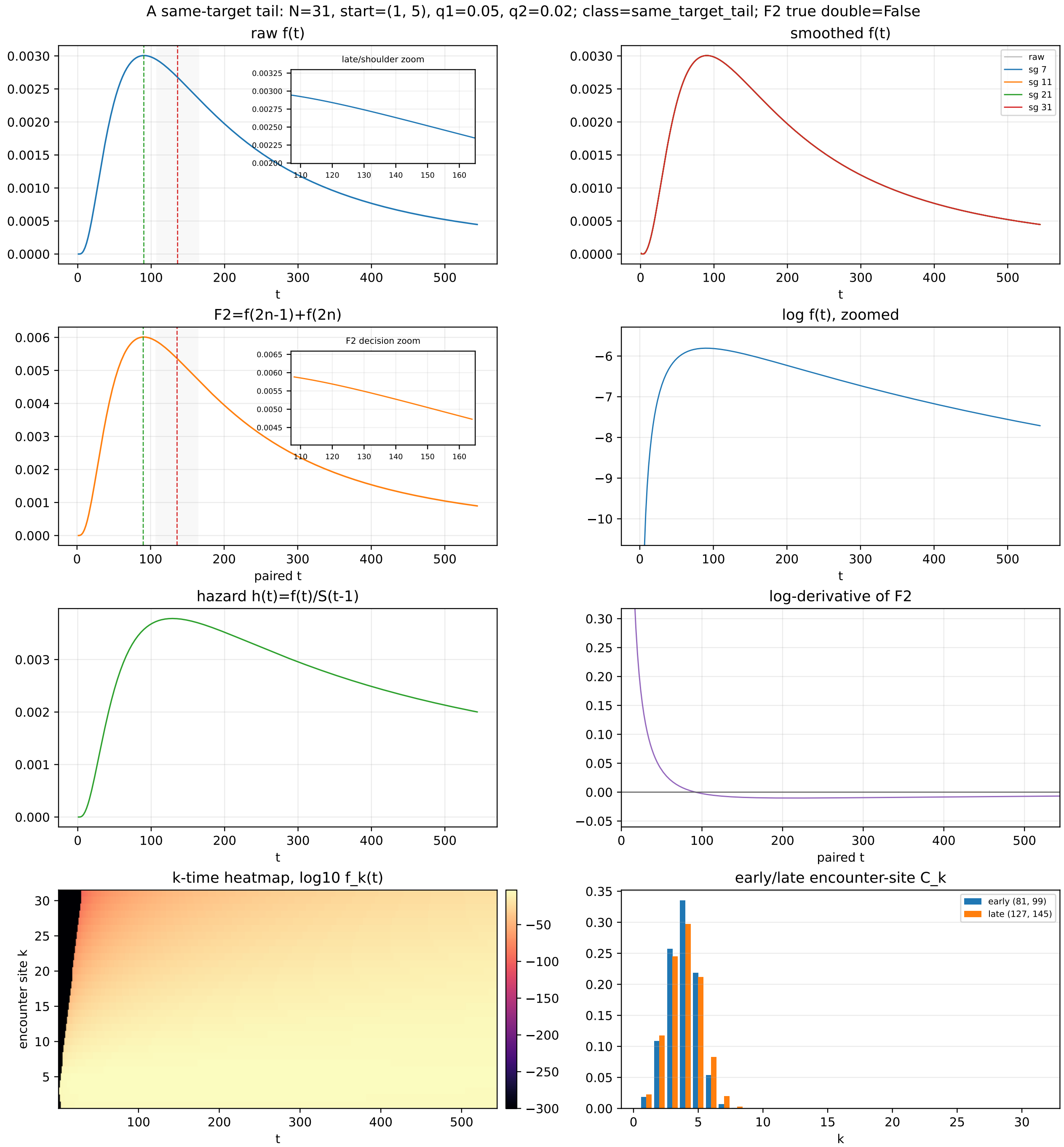

Figure 8: Full diagnostics for case A.

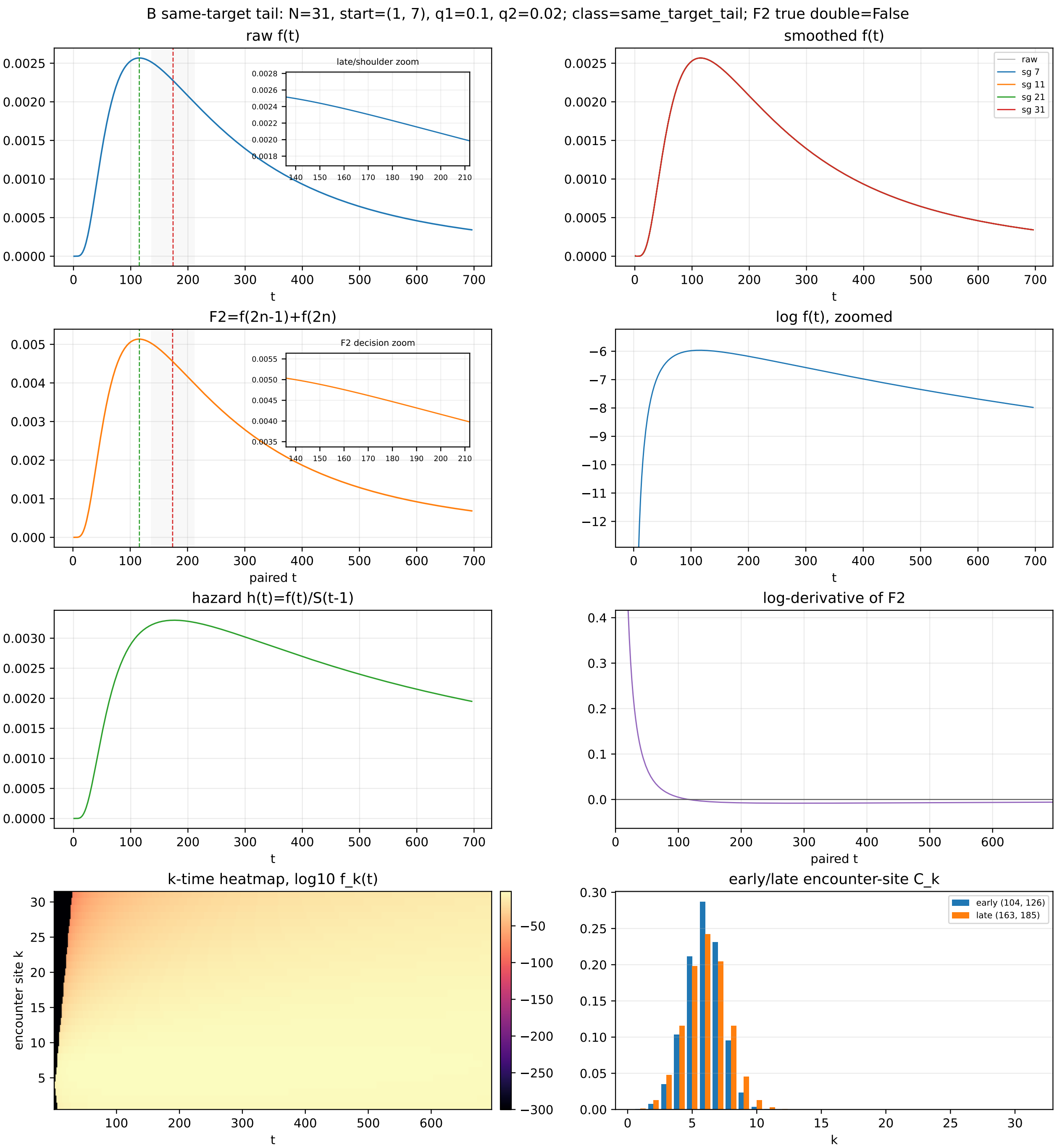

Figure 9: Full diagnostics for case B.

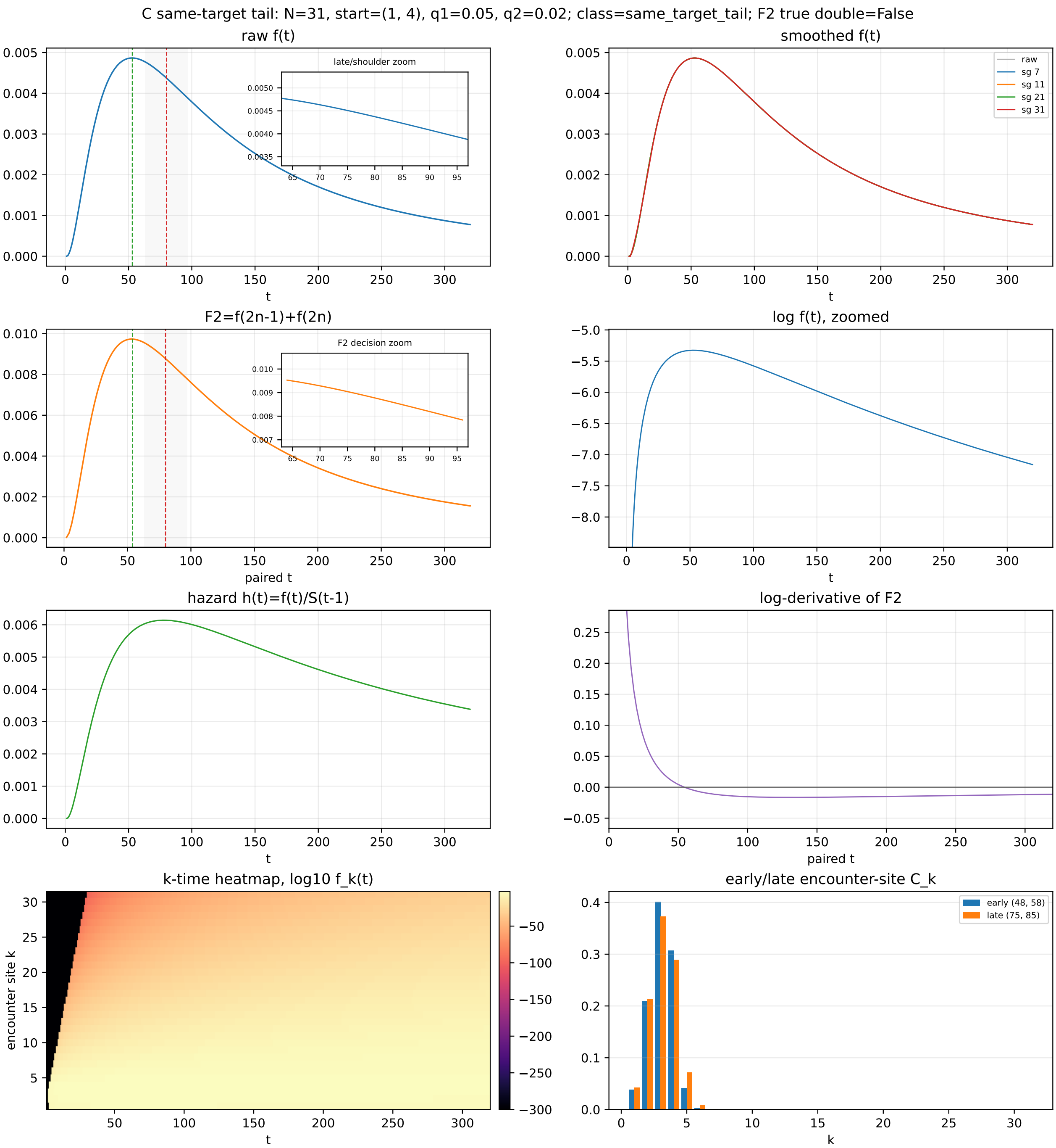

Figure 10: Full diagnostics for case C.

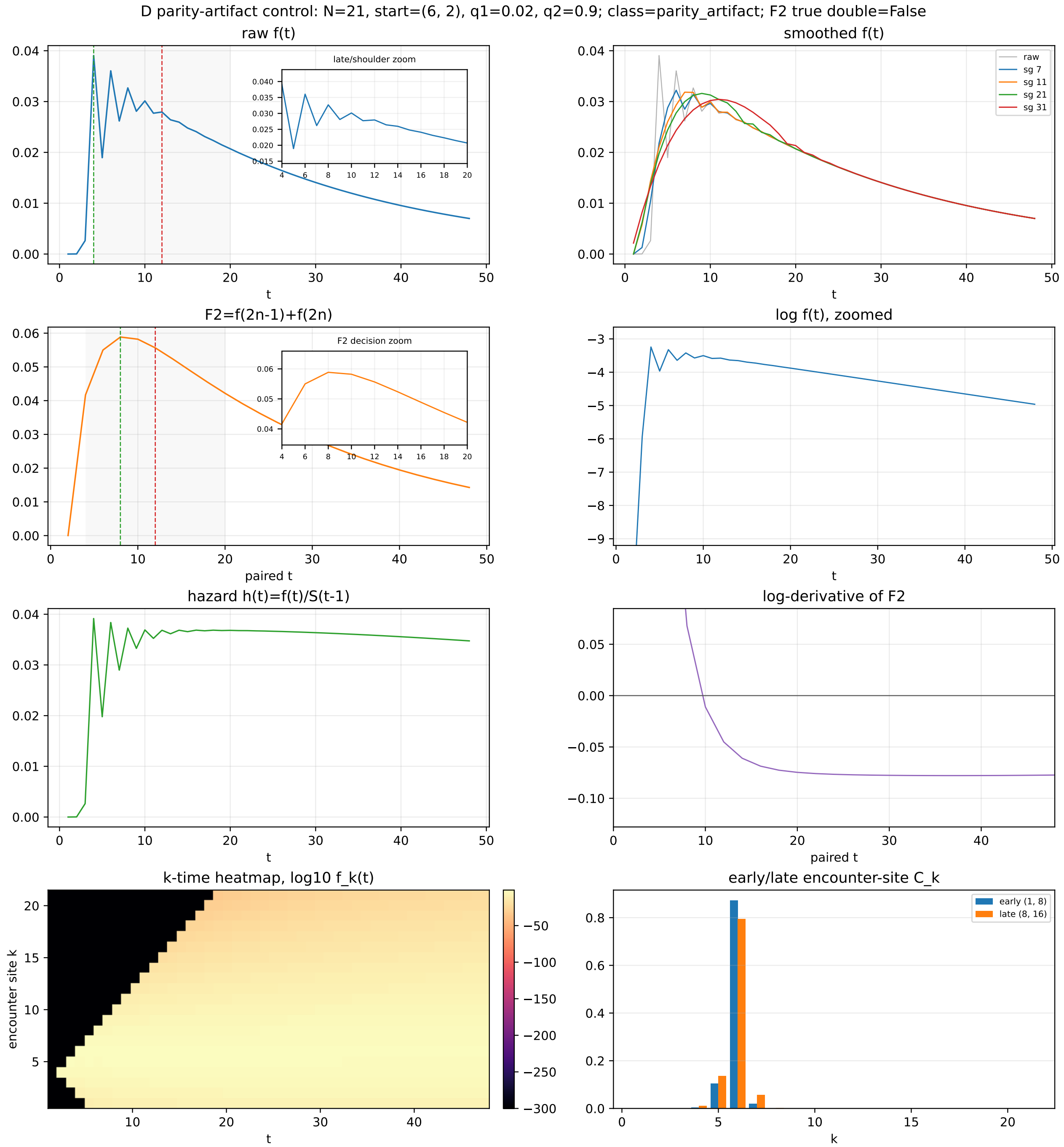

Figure 11: Full diagnostics for the Stage-1 negative control.

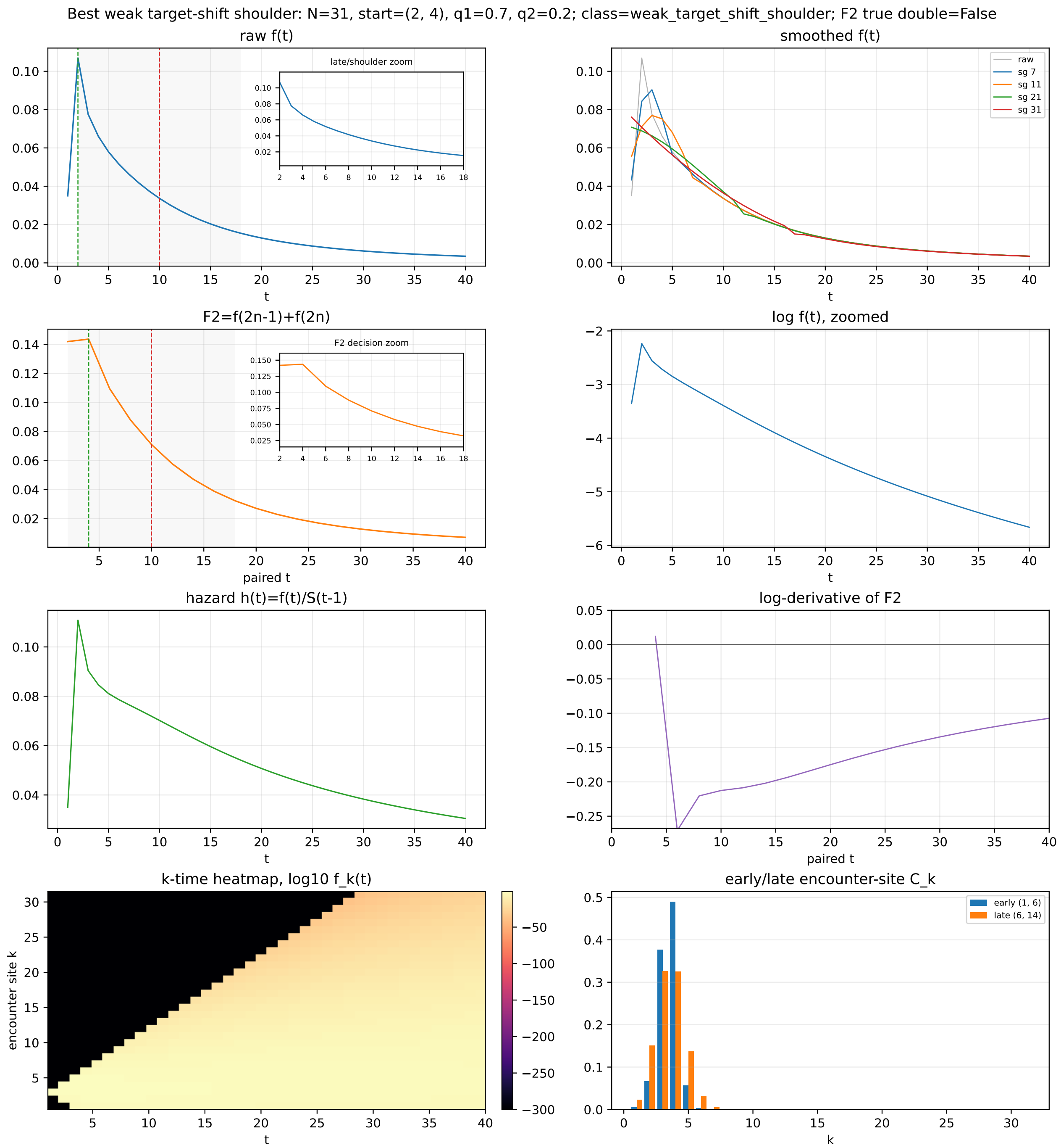

Figure 12: Full diagnostics for the best weak target-shift shoulder.

The tests used for the final package are

```
.local/encounter-py311/bin/python -m pytest \
  tests/test_encounter_search.py \
  tests/test_encounter_reflecting_mean_validation.py \
  tests/test_encounter_green_formula_comparison.py -q
```